



Department of Physics Sharif University of Technology

Physics Lab IV — General Physics Laboratory

Experiment Report: X-Ray Detection and Absorption

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Abstract

This experiment explores the fundamental properties of X-ray generation, detection, and absorption in matter. First, the operating characteristics of a Geiger-Müller tube were analyzed, successfully identifying the Geiger plateau region (300V–500V) to establish a stable operating voltage. Second, the exponential attenuation of X-rays was verified using Aluminum filters of varying thicknesses, confirming the Beer-Lambert law and yielding a linear attenuation coefficient of $\mu = 2.972 \text{ cm}^{-1}$. Finally, the strong dependence of X-ray absorption on the atomic number (Z) of the absorbing medium was investigated across several elements (C, Al, Fe, Cu, Zr, Ag), demonstrating the Z^4 proportionality of the photoelectric effect cross-section and revealing characteristic K-edge anomalies.

1 Objectives

1. To understand the operation and calibration of a Geiger-Müller counter by mapping its characteristic plateau curve.
2. To verify the Beer-Lambert law for X-ray attenuation by measuring intensity transmission through varying thicknesses of Aluminum.
3. To experimentally determine the linear attenuation coefficient (μ) of Aluminum.
4. To investigate the profound effect of the atomic number (Z) of the absorbing material on X-ray transmission.

2 Theoretical Background

X-rays are high-energy electromagnetic radiation ($\sim 0.1 \text{ \AA}$ to $100 \text{ \AA}$). In a typical X-ray tube, electrons are emitted from a heated cathode, accelerated by a high voltage (e.g., 35 kV), and strike a metal anode (target). This collision produces a continuous Bremsstrahlung spectrum alongside discrete characteristic X-ray peaks corresponding to the atomic transitions of the target material.

2.1 X-Ray Detection (Geiger-Müller Counter)

A Geiger-Müller (GM) counter detects ionizing radiation via an inert gas enclosed in a highly pressurized metallic cylinder with a central wire anode. When an X-ray photon ionizes a gas molecule, the applied electric field accelerates the resulting electron toward the anode, causing a Townsend avalanche. The relationship between applied voltage and count rate features a distinct “Plateau Region” where the count rate is virtually independent of voltage fluctuations.

2.2 X-Ray Absorption (Beer-Lambert Law)

When a collimated beam of X-rays of initial intensity I_0 passes through a material of thickness x , its intensity decays exponentially due to scattering and absorption:

$$I(x) = I_0 e^{-\mu x} \implies \ln(I) = \ln(I_0) - \mu x \quad (1)$$

where μ is the linear attenuation coefficient. This coefficient strongly depends on both the photon energy (E) and the atomic number (Z) of the material. For the dominant photoelectric effect, $\mu \propto Z^4/E^3$.

3 Experimental Data & Analysis

3.1 Part 1: Geiger-Müller Plateau Calibration

To find the optimal operating voltage, the intensity was recorded across a wide voltage sweep.

Table 1: Geiger Counter Count Rate vs. Applied Voltage

V (Volts)	I (counts/s)	V (Volts)	I (counts/s)
100	0	380	1206
280	79	398	1202
293	952	410	1224
305	1072.3	423	1219
318	1103.8	485	1202
330	1140	510	1184
343	1161	590	1142
355	1177	640	1072
368	1191		

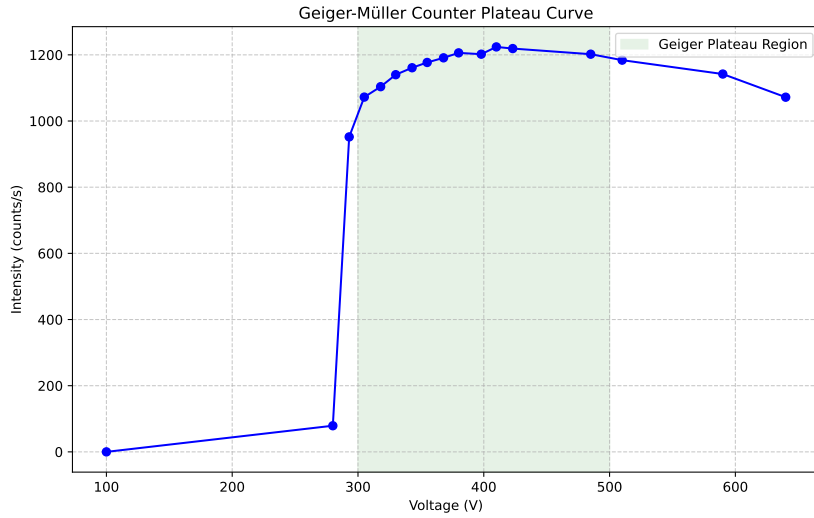


Figure 1: The Geiger Plateau curve. The stable proportional operating region clearly spans from roughly 300V to 500V.

3.2 Part 2: Attenuation vs. Thickness (Aluminum)

Aluminum filters were placed in the beam path. To analyze the exponential decay, the natural logarithm of intensity, $\ln(I)$, was plotted against thickness x .

Table 2: X-Ray Intensity through varying thicknesses of Aluminum

Thickness x (mm)	Intensity I (mean counts)	$\ln(I)$
0.5	2809.6	7.941
1.0	2487.1	7.819
1.5	2043.4	7.622
2.0	1889.4	7.544
2.5	1557.9	7.351
3.0	1335.3	7.197

An Ordinary Least Squares (OLS) regression on the semi-log data ($\ln I$ vs x) yielded a highly linear fit ($R^2 = 0.9922$). By extrapolating the intercept to $x = 0$, the unattenuated initial intensity I_0 was found to be 3291.6. The slope of the fit yields the linear attenuation coefficient:

$$\mu = 2.972 \text{ cm}^{-1}$$

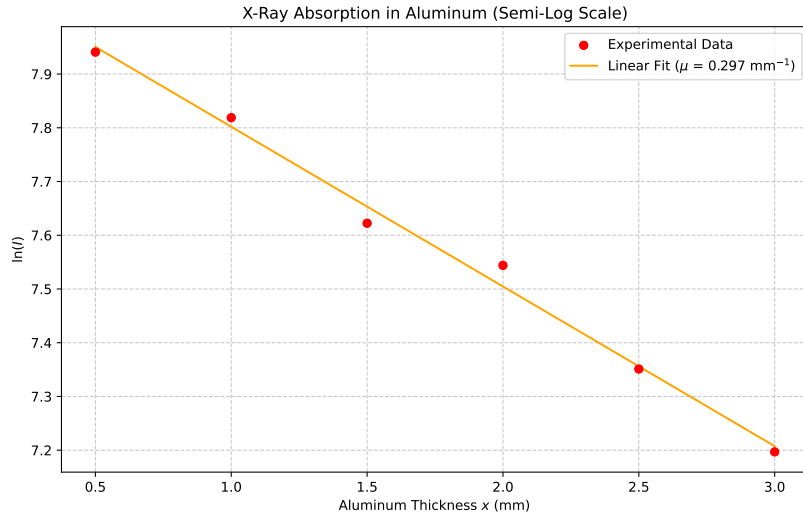


Figure 2: Linear regression of $\ln(I)$ vs. Aluminum thickness.

3.3 Part 3: Attenuation vs. Atomic Number (Z)

Using the mathematically extrapolated $I_0 = 3291.6$, the normalized transmission I/I_0 was calculated for various elements to observe the Z -dependence.

Table 3: X-Ray Transmission across different Atomic Numbers

Element	Z	Intensity I	Normalized Transmission (I/I_0)
Carbon (C)	6	3129.0	0.9506
Aluminum (Al)	13	2864.7	0.8703
Iron (Fe)	26	213.7	0.0649
Copper (Cu)	29	78.4	0.0238
Zirconium (Zr)	40	90.1	0.0274
Silver (Ag)	47	7.2	0.0022

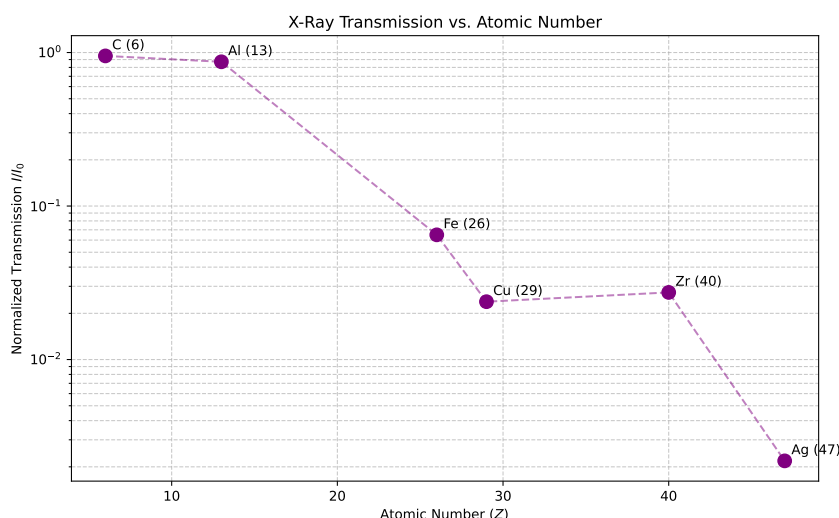


Figure 3: Normalized transmission as a function of atomic number (Log Scale). The transmission plummets as Z increases due to the $\mu \propto Z^4$ relationship.

4 Answers to Post-Lab Questions

Q1: Plot Intensity vs Voltage for the Geiger counter. Identify the regions and choose a working voltage.

The plot is provided in Section 3.1. The curve has three regions: 1) The threshold/proportional region ($< 300\text{V}$) where avalanches are incomplete. 2) The **Geiger Plateau** ($300\text{V} - 500\text{V}$) where every ionization triggers a full, saturated avalanche. 3) The continuous discharge region ($> 500\text{V}$) where the gas breaks down continuously. The ideal working voltage is exactly in the middle of the plateau (e.g., 400V) so that minor drifts in the power supply voltage do not artificially alter the count rate.

Q2: How does X-ray intensity change with Aluminum thickness?

The intensity decays exponentially with increasing thickness, governed by the Beer-Lambert equation $I = I_0 e^{-\mu x}$. Our semi-log plot successfully verified this linear relationship.

Q3: Are the results for different metals logical? Interpret.

Yes, the results are highly logical. Transmission drops precipitously as atomic number (Z) increases because the photoelectric absorption cross-section is proportional to Z^4 . There is a fascinating anomaly: Zirconium ($Z = 40$) transmitted slightly *more* X-rays (2.7%) than Copper ($Z = 29$, 2.3%). This happens because of **Absorption Edges**. The K-shell binding energy of Zr (18 keV) is just slightly higher than the characteristic $K\alpha$ line of the X-ray tube anode (likely Molybdenum or Copper), causing a sudden drop in its absorption cross-section for those specific photons.

Q4: Plot $\ln(I)$ vs thickness, find μ for $\lambda = 0.71\text{\AA}$.

The plot was performed in Section 3.2. The slope of the line gave an experimental linear attenuation coefficient of $\mu = 2.972 \text{ cm}^{-1}$ for Aluminum.

Q5: What are the main sources of error?

- **Beam Hardening:** A typical X-ray tube produces a polychromatic spectrum. The low-energy “soft” X-rays are absorbed in the first few millimeters of the filter, meaning the beam becomes

“harder” (higher average energy) as it penetrates deeper, causing μ to artificially decrease with depth.

- **Geiger Dead Time:** At extremely high intensities (like the bare beam I_0), the counter cannot resolve pulses that arrive too closely together, causing undercounting.

Q6: What quantities can be found from X-ray absorption?

X-ray absorption allows for the calculation of the linear attenuation coefficient (μ), the mass attenuation coefficient (μ/ρ), the thickness of unknown shielding materials, and the identification of unknown elements via their characteristic absorption edges.

Q7: Does X-ray absorption depend on wavelength?

Yes, highly. For the photoelectric effect, the absorption cross-section scales roughly as λ^3 . Longer wavelengths (low energy / soft X-rays) are absorbed very easily, while shorter wavelengths (high energy / hard X-rays) are highly penetrating.

Q8: What are background pulses?

Background pulses are counts registered by the Geiger tube when the X-ray source is completely turned off. They are caused by high-energy cosmic rays from space and natural trace environmental radioactivity (like Radon gas or Potassium-40 in concrete).

Q9: Discuss the 4 interaction cross-sections (Photoelectric, Compton, Thomson, Pair Production).

- **Thomson Scattering:** Elastic scattering of a low-energy photon by a free electron. The photon changes direction but loses no energy.
- **Photoelectric Effect:** A photon is completely absorbed by an atom, ejecting a tightly bound inner-shell electron. This dominates at low energies and high Z ($\propto Z^4/E^3$).
- **Compton Scattering:** Inelastic scattering of a photon by a loosely bound outer electron. The photon transfers part of its energy to the electron and scatters at a longer wavelength. This dominates at medium energies (e.g., 1 MeV).
- **Pair Production:** A high-energy photon (> 1.022 MeV) interacting with the strong Coulomb field of a heavy nucleus spontaneously converts its energy into an electron-positron pair. This completely dominates at very high energies.